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Associations of Self-Report Physical Function With Knee Strength and Knee Range-Of-Motion in Total Knee Arthroplasty
Possible Nonlinear and Threshold Effects

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ABSTRACT

This study aims to examine, in patients before and following a total knee arthroplasty (TKA), whether knee extensor strength and knee flexion/extension range-of-motion (ROM) were nonlinearly associated with physical function. Data from 501 patients with TKA were analyzed. Knee extensor strength was assessed preoperatively and 6 months postoperatively. Knee ROM and Short Form-36 (SF-36) physical function data were collected from each patient preoperatively, and at 6 and 24 months postoperatively. Knee strength was measured by handheld dynamometry and knee ROM by goniometry. Restricted cubic spline regression was used to examine possible nonlinear associations. At all assessment points, the associations between knee measures and function were not always linear. Some of the associations revealed distinct threshold points. These findings have potential clinical and research implications.

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Osteoarthritis (OA) of the knee is a major contributor to physical function limitations in older adults [1]. Although a total knee arthroplasty (TKA) can effectively relieve knee pain and restore function in patients with advanced stages of painful knee OA, 1 in 6 patients with TKA continues to have substantial physical function limitations [2]. Clearly, an in-depth understanding of the modifiable predictors of physical function is needed which, in turn, could assist the refinement of interventions to improve function.

Clinically, the common knee impairments before and following a TKA are reduced knee extensor strength and deficits in knee flexion and extension range-of-motion (ROM) [3]. Although previous studies [4–11] have implicated knee strength as an important predictor of physical function, the existing literature on the influence of knee ROM on physical function is mixed [5,6,8,11–16] and interpretation of this literature is hampered by small sample sizes in some studies [5,6,14,15]. Also, notably absent from most previous studies was an evaluation of nonlinear associations: many studies [4–15] – by their choice of standard regression or correlation analyses – have assumed that knee strength and ROM are related linearly to physical function.

Why is an examination of possible nonlinear associations important? We offer 3 reasons. First, the concept of nonlinear associations has biological plausibility [17] and empirical support [8,18–21] from previous studies not limited to patients with TKA. Second, a failure to identify nonlinear associations may produce misleading null results [22] and lead to the conclusion that a given knee measure is not predictive of physical function. Third, if the association between knee measures and physical function were indeed nonlinear and that clinical strength and ROM thresholds could be identified, these thresholds could have important clinical and research ramifications [22].

In summary, developing effective interventions depends on understanding both *whether* and *how* knee strength and ROM influence physical function. Because studies examining nonlinear associations are surprisingly sparse in the TKA literature, we initiated the present study to examine whether knee extensor strength and knee ROM were nonlinearly associated with self-report physical function in a large group of patients before and following a TKA.

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Materials and Methods

Settings and Participants

We conducted a retrospective analysis of 836 patients aged 50 years or older who underwent a primary TKA for knee OA performed by 3 high-volume surgeons at Singapore General Hospital, Singapore, from 3 January 2006 to 29 January 2009. To avoid potential confounding effects from a contralateral TKA operation, we excluded patients if they underwent a contralateral TKA within the 24 months before or after their index TKA ($n = 199$). We also excluded patients who (i) had a history of stroke or other neurological disorders ($n = 30$), (ii) had a history of lower limb fracture ($n = 12$), (iii) had previously undergone a hip arthroplasty or high tibial osteotomy ($n = 26$), (iv) had previously undergone a unicompartmental knee arthroplasty (UKA) on their index knee ($n = 4$), or (v) died in the first 6 months after a TKA ($n = 14$), (vi) had injurious falls in the first 6 months after a TKA ($n = 9$), (vii) had significant back, hip, or ankle pain ($n = 16$), or (viii) developed new medical conditions (e.g., stroke and dementia) or surgical complications prior to the first follow up session ($n = 25$). Thus, the sample for analysis comprised the remaining 501 patients who underwent a preoperative evaluation within 5 weeks prior to their operation. Follow-up assessments were conducted approximately 6 and 24 months after the operation. All data – preoperative and postoperative – were collected by physical therapists and entered into an electronic database per routine practice policies of our institution. Institutional review board approved the study.

Demographic Variables

The demographic information collected in this study included age (in years), gender, ethnicity (Singaporean Chinese, as compared with others), and education (1 = primary, 2 = secondary, 3 = tertiary). Data about comorbidities were retrieved from patients' medical records by a research assistant using a checklist modeled after the Self-Administered Comorbidity Questionnaire [23]. Patients' height and weight were also obtained and body mass index (BMI) was calculated as the ratio of weight to squared height (kg/m^2).

Outcome Variable

At each assessment point, patients were interviewed in either English or Mandarin using the Short Form-36 [24], of which we used the physical function subscale as the outcome variable. The SF-36 physical function subscale ranges from 0 to 100, with higher scores representing better physical function. The English and Chinese versions of the SF-36 have been previously validated for use in Singapore [25].

Knee Extensor Strength

At the preoperative and 6-month postoperative assessments, a hand-held dynamometer (Lafayette, IN, USA) was used to measure isometric knee extensor strength. Knee extensor strength was measured with the patients in supine position, and the knee was positioned in approximately 30° flexion by a firm wedge. The dynamometer force pad was placed just proximal to the ankle joint and knee extensor strength was quantified in kilogram force. All patients performed 2 maximal trials for 3 to 5 s with a 30-s rest interval. Additional measurements were taken if the patient reported a failure to achieve maximum effort. The higher of two valid trials was recorded and normalized as a direct percentage of body weight (%BW) because we believe that this index of knee strength made intuitive sense to our patients. Although we [26] and others [27] have shown, using allometric analyses, that the association between muscle force generation and body weight is not a strict 1:1 ratio, analyzing our data using allometrically-scaled knee extensor strength did not qualitatively alter our study conclusions (data not shown). Also of interest, we performed strength testing in the supine position – and not in the seated position – with the intent to reduce the influence of the rater's strength on the patient's measurements [28]. For the supine knee strength measurements, previous studies have demonstrated good test–retest reliability (intraclass correlation coefficients, 0.80 to 0.86) and concurrent validity with Biodex dynamometry measurements [28,29].

Knee Flexion and Extension ROM

At all assessment points, a large standard goniometer (Jamar, Clifton, NJ, USA) was used to measure passive knee flexion and extension ROM. Although passive ROM may not reflect functional ROM – that is, the knee ROM achieved by the patients during daily physical activities – we measured passive knee ROM to facilitate comparisons with most previous studies [8–12,14,16]. In our study, knee flexion ROM was measured with the patients in supine position. With the assistance of a belt which was hooked around the distal tibia and fibula, patients were asked to slide their heels toward the buttocks, pull on the belt, and flex their knees maximally. Of interest, this action mimics a knee flexion stretching exercise which purports to increase passive knee flexion ROM [30]. Knee extension ROM was measured with the patients in supine position with the heel elevated on a firm wedge. Two sets of flexion and extension ROM measurements were taken, and the higher ROM measurement was recorded. For the knee ROM measurements, one previous study in patients with knee OA has demonstrated good test–retest reliability (intraclass correlation coefficients, 0.95 and 0.85 for flexion and extension ROM, respectively) [31].

Statistical Analysis

We used descriptive statistics to characterize the study sample: we used means (SDs) for continuous variables and percentages for categorical variables. Analysis of variance (ANOVA) for repeated measures was used to test for change over time in SF-36 physical function and measures of knee strength and ROM.

We used separate multivariable regression models to examine the association between knee measures and SF-36 physical function at all time points, adjusting for age, gender, BMI, and the number of comorbidities. To explore nonlinear associations, we modeled the knee strength and ROM measures as restricted cubic splines with three knots located at the 10th, 50th, and 90th percentiles of the predictor distribution [22,32,33]. To assess nonlinearity, we used an *F*-test [34]. Of note, restricted cubic splines are smoothing functions which make no *a priori* assumptions about the shape of the curve and hence, allow us to flexibly model the association between the various knee measures and physical function [33]. Knots – locations where the piecewise spline functions join – do necessarily reflect inflection points of the curve, and it has been observed that the results from restricted cubic spline regression are usually insensitive to knot locations [33,34].

In our study, 13% of patients had missing data at the 6-month assessment (4% were missing data only on knee strength, 4% were missing data on both knee strength and ROM, and 5% were missing data on all 6-month variables). At the 24-month assessment, 13% of patients had missing data and data from a further 6% of patients were unusable because the patient developed other illnesses (5%), or underwent other operations either to their lower limb or spine (1%). Accordingly, to correct for potential selection bias due to missing data [35], we used a flexible additive regression model to impute missing predictor values [36], incorporating all demographic, predictor, and outcome variables [37]. In these analyses, we assumed that data were

missing at random. We generated 50 imputed data sets, replicated the analyses on each data set, and used Rubin's rules [38] to consolidate the regression coefficients and standard errors.

All statistical analyses were done with R software, version 2.15.0 (R Foundation, Vienna, Austria). Statistical significance was determined at the 2-sided 0.05 level.

Results

Sample Characteristics

Table 1 summarizes the participants' descriptive characteristics, while Table 2 provides the descriptive statistics for SF-36 physical function and knee measures. The patients were predominantly female (82%) and were on average moderately overweight (mean (SD) BMI, 28 (4.6) kg/m^2). Improvement in SF-36 physical function generally occurred in the first 6 months following a TKA, after which the scores appeared to plateau. Specifically, mean SF-36 physical function scores were 39 (SD = 23) points preoperatively, 67 (SD = 18) points at 6 months, and 68 (SD = 18) points at 24 months. Repeated measures ANOVA indicated that this postoperative improvement was statistically significant ($P<0.001$). With respect to the knee measures, knee extensor strength increased from preoperative levels by around 40% at the 6 month postoperative assessment (18%BW vs. 26%BW; $P<0.001$). Knee flexion ROM did not improve across the time points whereas knee extension ROM improved with time, from 6.8° preoperatively to 1.1° at 24 months.

Knee Extensor Strength

Table 3 shows the results of the tests for overall and nonlinear multivariable associations between SF-36 physical function and measures of knee strength and ROM. Preoperatively and 6 months postoperatively, knee extensor strength was significantly associated with SF-36 physical function ($P<0.001$), with evidence for nonlinearity (P's = 0.01) (Table 3). Specifically, a steep rise in SF-36 physical function with increasing knee strength was observed up to an approximate threshold value of ~25%BW, above which there was a more gradual rise (Fig. 1A and 1D).

Knee Flexion and Extension ROM

Knee flexion ROM was significantly associated with SF-36 physical function preoperatively and 24 months postoperatively (Table 3). At the 24-month postoperative assessment, we observed no appreciable association between knee flexion ROM and SF-36 physical function below 120° of knee flexion; however, the flexion–function association increased sharply at greater degrees of knee flexion (>120°) (P-value for nonlinearity = 0.03; Fig. 1G).

Knee extension ROM was significantly associated with SF-36 physical function across all time points (Table 3). At the 24-month postoperative assessment, the association between knee extension ROM and SF-36 physical function was significantly nonlinear (P = 0.01), approximating an inverted V-shape with a distinct inflection point (peak) at full knee extension (0°) (Fig. 1H). Put otherwise, compared patients with full knee extension, patients with knee flexion contracture (extension ROM>0°) or knee hyperextension (extension ROM <0°) reported lower SF-36 physical function.

Discussion

The purpose of this study was to describe the association of SF-36 physical function with knee extensor strength and knee flexion/extension ROM in a large sample of patients before and following a TKA, with a focus on possible nonlinear associations. Our results suggest that at all assessment points, the associations were not always linear and some of the associations departed from linearity to reveal

Table 1
Demographics and Patient Characteristics (n = 501).

Characteristic	Value
Age, years	67.9 ± 7.8
Male gender, no. (%)	92 (18%)
Singaporean Chinese, no. (%)	439 (88%)
Height, m	1.53 ± 0.07
Body mass, kg	65.62 ± 12.1
BMI, kg/m^2	27.9 ± 4.6
Overweight or obese, no. (%)	428 (85%)
Education, no. (%)	
Primary or less	357 (76%)
Secondary	90 (19%)
Tertiary	22 (5%)
History of TKA[a]	75 (15%)
Comorbidities, no. (%)	
Hypertension	336 (67%)
Diabetes	105 (21%)
Cardiac	78 (15.5%)
≥2 comorbidities	245 (49%)

Note: Values reported were mean ± SD unless stated otherwise.
Abbreviations: TKA = Total knee arthroplasty; BMI = body mass index.
[a] History of TKA on the non-index knee more than 2 years before the index operation.

Table 2
Postoperative Recovery of Knee Measures and SF-36 Variables.

Variables	Preop (n = 501)	6 Months	24 Months	n at 6 Months/n at 24 Months	P-Value[a]
SF-36 Physical function	38.8 (22.6)	66.6 (17.9)	67.8 (17.8)	471/417	<0.001
Knee variables					
Flexion ROM, deg	116.1 (18.1)	113.2 (14.1)	116.7 (13.9)	453/411	<0.001
Extension ROM, deg	6.8 (8.7)	3.6 (5.1)	1.1 (4.7)	453/411	<0.001
Extensor strength, %BW	18.3 (8.4)	25.5 (8.3)	–[b]	435/–[b]	<0.001

Note: Values reported were mean (SD) unless stated otherwise.
Abbreviations: SF-36 = Medical Outcomes Study 36-Item Short-Form Health Survey; ROM = range of motion; BW = body weight.
[a] P-values derived from repeated-measure ANOVA. With the exception of knee flexion ROM, all measures improved from the preoperative assessment to the 6-month postoperative assessment.
[b] Knee extensor strength data were not available at the 24-month postoperative assessment.

Table 3
Tests for Overall and Nonlinear Multivariable Associations Between SF-36 Physical Function and Measures of Knee Strength and ROM.

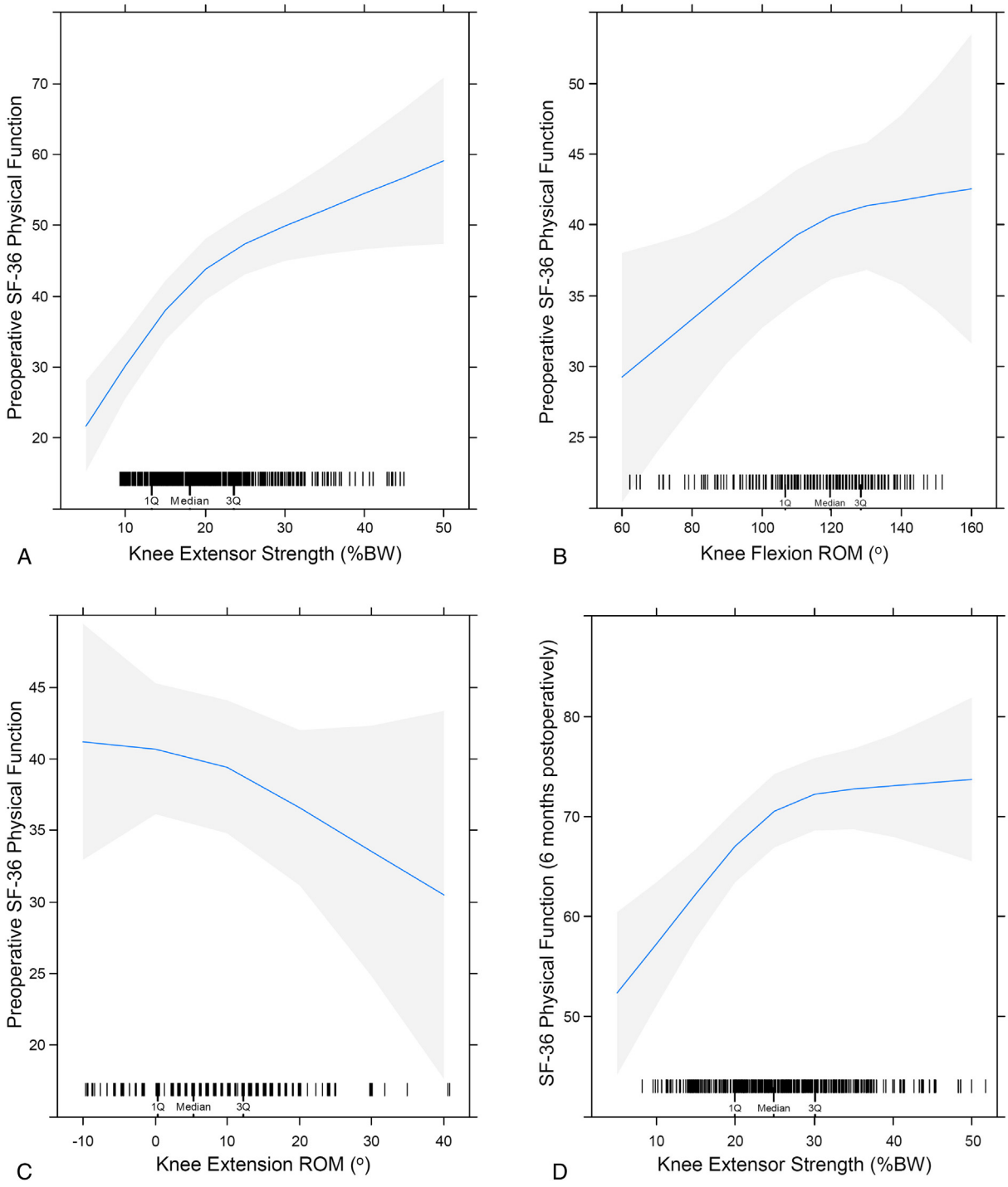
	Preoperative Assessment		6-Month Postoperative Assessment		24-Month Postoperative Assessment	
Variables[a]	F-value	P-value	F-value	P-value	F-value	P-value
Knee strength						
Overall association	34.27	<0.001	13.54	<0.001	–[b]	–[b]
Nonlinear association	6.65	0.01	6.41	0.01	–[b]	–[b]
Knee flexion ROM						
Overall association	4.02	0.02	0.51	0.60	4.61	0.01
Nonlinear association	0.68	0.41	0.17	0.68	4.63	0.03
Knee extension ROM						
Overall association	3.11	0.04	4.93	<0.01	4.14	0.02
Nonlinear association	1.13	0.28	2.49	0.11	6.38	0.01

Note: All analyses were adjusted for age, gender, body mass index, and number of comorbidities. The 6-month and 24-month analyses were based on 50 multiply imputed data sets.
Abbreviations: SF-36 = Medical Outcomes Study 36-Item Short-Form Health Survey; ROM = range of motion.
[a] Knee strength and ROM were modeled as three-knot restricted cubic splines.
[b] Knee strength data were not available at the 24-month postoperative assessment.

distinct threshold points. To our knowledge, these findings have not been previously described in patients with TKA.

In our sample, knee extensor strength emerged as a robust predictor of physical function (Table 3) — a finding consistent with that of many prior studies [4–6,14,15]. Unlike previous studies, however, our results are more nuanced: we observed evidence of nonlinearity in the association between knee extensor strength and physical function, with a stronger increase in SF-36 scores within the lower versus the higher levels of knee strength (strength threshold ~25%BW; Figs. 1A and 1D). Although we found no studies sufficiently similar to ours to make direct comparisons, our findings appear to support the theory by Buchner and de Lateur [17] which predicts that the association between muscle strength and physical function is nonlinear, with a strength threshold below which the muscle strength and physical function are closely related, but above which the association attenuates. Unfortunately, we did not measure knee strength at the 24-month postoperative level which would have further elucidated the precise functional impact of knee extensor strength. Nevertheless, our findings support the potential value of exploring this issue because if strength thresholds were to truly exist, they could serve as objective rehabilitation goals.

When we used a spline function to model knee flexion ROM, its overall association with physical function was significant preoperatively and 24 months postoperatively — but not 6 months postoperatively. Although somewhat unexpected, the latter null results are not without precedence: several studies have reported



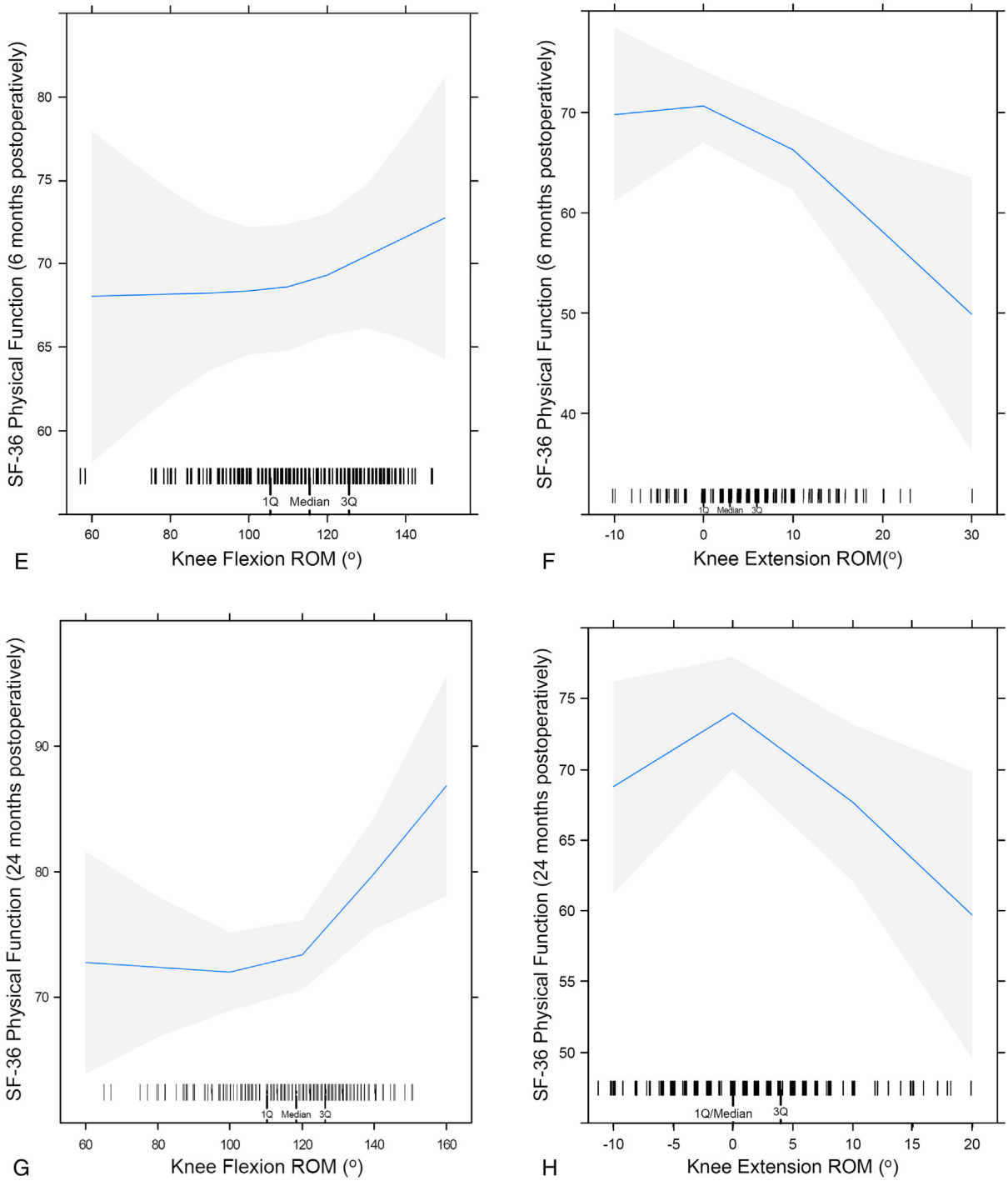


Fig. 1. Associations of SF-36 physical function with knee extensor strength (A and D), knee flexion ROM (B, E, and G), and knee extension ROM (C, F, and H) preoperatively (A to C), 6 months postoperatively (D to F), and 24 months postoperatively (G and H). Shaded regions represent 95% CIs for the spline regression estimates. The median, first quartile (1Q), third quartile (3Q), and all observed values of the knee impairment variables are indicated by the short vertical lines above the x-axes.

that knee flexion ROM correlated significantly with self-report function preoperatively [7,13] but less so 6 to 12 months postoperatively [12,13,15,16]. Beyond these findings, there are no clear explanations for our results. At the 24-month postoperative assessment – a time point at which physical activity levels were reportedly higher than preoperative levels [9] – we found no appreciable relationship between knee flexion ROM and SF-36 physical function up to a threshold angle of ~120°, above which physical function incrementally increased with increasing flexion ROM (Fig. 1G). Reviewing the literature, our findings are generally consistent with those from 2 recent investigations showing a pronounced, positive impact of higher knee flexion ROM (>125°, Ref. [10], and >130°, Ref. [9]) on self-report stair climbing ability or overall physical function 4 years post knee arthroplasty. However, our study has several methodological advantages over past work, the biggest of which is the use of restricted cubic spline regression to avoid the arbitrariness and statistical limitations [32,39] associated with categorizing patients into subgroups. For these reasons, we believe our results complement and extend the evidence for a nonlinear association between knee flexion ROM and physical function in patients who had undergone a TKA at least 2 years ago.

On a practical level, it seems natural to question the usefulness of improving patients' knee flexion ROM beyond 120°, given that most activities of daily living – for example, stair climbing and standing from sitting in a chair – are usually performed with the knees flexed, at a maximum, to 110° [40]. We offer 2 justifications. At the simplest, certain physical tasks that the SF-36 enquires – namely, kneeling and lifting – do demand much greater knee flexion [41]; hence, improving flexion ROM beyond 120° would be expected to confer additional functional benefits. The other justification derives from work by Escalante et al [21] which demonstrated, in older adults, a nonlinear, J-shaped association between knee flexion ROM and the odds of walking slowly, with the odds increasing sharply when knee ROM fell below 140°. Together, these findings suggest that improving flexion range beyond 120° may be necessary for optimal physical functioning.

In our study, we observed significant associations for knee extension ROM at all time points (Table 3), with reduced physical function amongst patients with greater knee flexion contracture (extension ROM>0°) (Figs. 1C, 1F, and 1H). Of interest, at the 24-month postoperative assessment, the peak of the spline curve was full knee extension (0°) (Fig. 1H). Although the biological plausibility of these findings are supported by clinical observations [42,43] and experimental studies [44,45] which have highlighted the importance of full knee extension ROM in normal knee biomechanics, previous TKA studies relating knee extension ROM to physical function have yielded conflicting results [8,11–13]. Of these studies, inferences from 2 previous studies with null results [12,13] were hindered by the use of linear, correlational analyses in samples which included patients with knee hyperextension. Interestingly, had we modeled knee extension ROM linearly in our analyses, we would have also obtained a non-significant result ($P = 0.14$) (data not shown). In contrast, in the one study which specifically excluded patients with knee hyperextension [11], a strong, linear relationship between knee extension ROM and physical function was observed. Finally, our nonlinear findings are in accordance to the retrospective study by Ritter et al [8] on 3672 patients. In that study, the odds ratios for poor postoperative Knee Society Function scores (<70 points) comparing patients with severe knee hyperextension (extension ROM ≤ −10°) or flexion contracture (ROM ≥ 20°) with those with normal knee extension (between −9° and 5°) were 1.9 and 6.7, respectively. These findings, when combined with ours, suggest that knee extension ROM is an important predictor of physical function and that previous null findings may have resulted from not considering nonlinear effects. Future studies should explore this possibility in their analyses.

Our study has implications. First, by demonstrating the nonlinear effects of knee ROM on physical function, our findings help reconcile some of the previously reported null and equivocal findings [8,11–13] and they counter previous nihilism [14–16] toward improving knee ROM in patients with TKA. Relatedly, given that the majority of our patients were unable to flex more than 120° post TKA (Table 2), a finding also reported by other studies [6,9,12,14,46], our findings suggest that exploring how best to improve knee flexion ROM, be it via better implant design or rehabilitation planning, may constitute a worthy endeavour. Second, because our use of cubic splines provides a more visual and nuanced description of the associations between knee measures and physical function, our results, if verified in future experimental studies, may be useful in (i) treatment goal setting, (ii) patient education, and (iii) identification of patients toward whom greater rehabilitation efforts should be directed. Also, from a research perspective, the same information could potentially assist in treatment planning, patient subgrouping, and sample size estimation [22].

Study Limitations

Our study has limitations. First, our study comprised a preponderance of Asian women. Because of possible ethnic and gender differences in knee ROM requirements [47,48], the generalizability of our results to other samples may be limited. Second, whilst we used a self-report measure of physical function to facilitate data collection, we acknowledge that performance-based measures of physical function are necessary because they provide important, complementary information about functional status [49] and because the precise association between knee strength and ROM measures and a given physical task (e.g. stair climbing and walking) is dependent on the movement demand of the task itself [18]. Third, our cross-sectional results may be affected by reverse causation and we also cannot exclude residual confounding from unknown or unmeasured factors.

In conclusion, in patients with TKA, the associations between self-report physical function and measures of knee strength and ROM were certainly less straightforward than previously assumed. Although these results await experimental verification, we found that having (i) knee extensor strength exceeding 25%BW, (ii) knee flexion ROM beyond 120° and (iii) full knee extension ROM were associated with optimal physical functioning. This information can help guide clinical practice as well as research planning.

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